SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: Artificial Intelligence **COURSE CODE:** CSA17

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively.*(BL3)*

Assessment Tool Weightage: 35%

QUESTIONS: 3

Duration : 1 Hour
Total Marks:30

Annexure A

Constraint-Based Problem Solving

Code of Conduct:

I M. Dhanushkumar / 192525318 *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: M Dhanush Kumar

Annexure A

Constraint-Based Problem Solving

1. A hospital needs to assign 3 doctors (D1, D2, D3) to 3 shifts (Morning, Afternoon, Night) such that:

   Constraints:

   i. D1 cannot work Night shift
   ii. D2 must work before D3 (Morning < Afternoon < Night)
   iii. Only one doctor per shift
   iv. D3 cannot work Morning
   v. All doctors must be assigned exactly one shift

   Apply Backtracking Search to find a valid assignment with all intermediate steps and constraint checks. State the final valid schedule

2. A robot operates in a grid environment (5×5). It must move from Start (S) to Goal (G). Obstacles block certain paths.

   Constraints:

   i. Movement allowed: Up, Down, Left, Right
   ii. Cost of each move = 1
   iii. Cannot pass through obstacles
   iv. Use Manhattan Distance heuristic

   Using above constraints and BFS Search Algorithm Show step-by-step node expansion and priority queue updates and determine the optimal path and cost.

3. An autonomous rescue robot is deployed in a disaster-affected building to locate and rescue survivors. The robot operates in a partially observable environment where some paths are blocked, and it must make decisions with limited information.

The building is represented as a grid of rooms. The robot starts at position S and must reach a survivor located at G.

Constraints:

- The robot can move up, down, left, right
- Some cells are blocked (obstacles) and cannot be traversed
- The robot has limited sensing capability (can only observe adjacent cells)
- Each movement has a cost of 1, but entering risky zones has an additional cost of 2
- The robot must minimize total cost while ensuring safe navigation
- The robot must update its knowledge dynamically (online search scenario)

Tasks:

a) Analyze all the given constraints and explain how they affect the robot's decision-making process.

b) Develop a solution approach using an appropriate search strategy (e.g., Uniform Cost Search / Online Search Agent). Clearly explain the steps involved.
c) Demonstrate how the robot applies AI concepts such as:

- Intelligent agents
- Rationality
- Environment type

d) Justify why your chosen approach is the most suitable for solving this problem under the given constraints.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	25%	Clearly defines the problem and identifies all relevant constraints accurately	Identifies most constraints with minor gaps	Partial understanding; misses key constraints	Fails to identify constraints correctly
Constraint Analysis	25%	Analyzes constraints effectively and prioritizes them logically	Provides reasonable analysis with some prioritization	Limited analysis; weak prioritization	No meaningful analysis or prioritization
Solution Development	25%	Develops optimal and feasible solutions satisfying all constraints	Develops workable solutions with minor limitations	Solutions partially satisfy constraints	Solutions do not meet constraints
Application of Concepts	15%	Effectively applies appropriate concepts, methods, or tools	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply relevant concepts
Justification	10%	Provides strong justification for decisions with logical reasoning	Provides justification with some clarity	Weak or unclear justification	No justification provided

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CO1-AT2-Constraint Based Problem Solving

1) Hospital Doctor shift Assignment using Backtracking Search.

Problem:- Assign three doctors (D1, D2, D3) to morning, Afternoon and Night Shifts while satisfying all the given Constraints.

Constraints:-

1. D1 Cannot work the Night shift
2. D2 must work before D3
3. Only one doctor can work per shift.
4. D3 cannot work the Morning shift.
5. Each doctor must be assigned exactly one shift

Find schedule

Doctor	shift	Status
D1	Morning	Valid
D2	Afternoon	Valid
D3	Night	Valid

Result:-

The final Valid Schedule is D1 - morning, D2 - Afternoon And D3 - Night. All constraints are Satified.

2) Robot Navigation in a 5×5 Grid using BFS

Problem: A robot must move from start (S) to Goal (G) in a 5×5 grid without crossing obstacles.

Each movement costs 1.

Constraints

- The robot can move up, Down, Left and right.
- Each move costs 1.
- The robot cannot cannot Pass through obstacles.

$$h(n) = |x_G - x_n| + |y_G - y_n|$$

Procedure:

1. Insert the start node into the queue.
2. Expand S and check its neighbouring cells.
3. Remove cells that Contain obstacles or lie outside the Grid.

4. Insert all valid, unvisited neighbours into the queue.
5. Continue expanding nodes in FIFO order.
6. When G is reached, trace back the path using parent nodes.
7. Since each movement costs 1, the total costs equals the number of moves.

Result:-

BFS finds an optimal shortest path when every move has equal cost. However, the uploaded question does not provide the exact positions of S, G and the the obstacles in its extract text, So the exact node expansion, Path and Path and numberical cost cannot be determined.

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3) Autonomous Rescue Robot in a Disaster-Affected Building

a) Analyze the constraints

Constraint — Effect on the robot

Constraint	Effect on the robot
Four-direction movement	It can move up, down, left and Right
Obstacles	It must avoid blocked cells
Limited Sensing	It can observe only Adjacent Cells
Movement Cost	Each normal move costs 1:
Risky Zones	Entering a risky cell costs 3 in total (1+2 extra).
Dynamic Knowledge	It must update its map when it discovers new information.

b) Solution using uniform cost search and olive search.

Steps:

1. Start at S with cost 0.
2. Observe Adjacent cells using sensors
3. Identify available Paths, obstacles and risky Zones
4. Assign cost 1 to a normal move and costs to risky zones.
5. Expand the lowest-cost position first.

6. ~~Expand the map when new obsta~~

6. Expand the lowest-cost position first.

7. Update the map when new obstacles or risky zones are discovered.

(c) Application of AI Concepts

1. Intelligent Agent: the robot senses the surroundings processes information and selects actions to reach the survivor.

2. Rationality: A rational robot chooses actions that help it reach a while minimizing cost, avoiding obstacles and reducing risk using the information available.

Results- Combining online search with UCS-based cost evaluation helps the robot to reach the survivor while avoiding obstacles, considering risky zones and minimizing total travel cost.